



Health prognosis via feature optimization and convolutional neural network for lithium-ion batteries

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ABSTRACT

With the rapid expansion of the electric vehicle market, the demand for lithium-ion batteries (LIBs) is exploding. The state of health (SOH) of LIBs is receiving more widespread attention, which is the key parameter for battery health management. This paper proposes a SOH estimation method for LIBs via feature optimization and convolutional neural network (CNN) to reduce the information redundancy of existing multiple features, aimed at leveraging multiple sources of features while optimizing their combination to minimize redundancy. Firstly, multiple features are extracted from different perspectives, including electrical, thermodynamic, and electrochemical properties, to comprehensively characterize the aging of batteries. Secondly, we construct a SOH estimator based on principal component analysis (PCA) with CNN (PCA-CNN). Finally, the dimension of features is optimized with a simulated annealing algorithm (SA) under the mean-variance objective function. Moreover, Comparative experiments are conducted on the Oxford dataset for validation. The results demonstrate the effectiveness of the proposed multi-feature description method in terms of accuracy and smoothness. Compared to traditional CNN methods and fixed-dimension PCA-CNN, this estimation approach significantly improves performance, showing more than 20% and 30% increases in the key metrics of MAE and RMSE, respectively. This study successfully optimized feature combinations to reduce redundancy within the feature set while enhancing the accuracy of SOH estimation.

1. Introduction

In recent years, due to global warming and air pollution caused by traditional vehicle transportation, extensive research has been devoted to the development of electric vehicles (EVs). EVs have gradually replaced conventional vehicles due to the significant advantages of their equipped lithium batteries (LIBs), i.e., high energy density, low cost, excellent recyclability, and long life (Zhang and Li, 2022). However, as the cumulative driving range of EVs increases, the LIBs will gradually age during the charge-discharge cycles, which will directly affect the remaining useful life of LIBs. In addition, if the deterioration of battery state-of-health (SOH) is not detected in time, it may further affect the safety and reliability of EVs (Xiong et al., 2018), (Meng et al., 2023).

Consequently, the accurate estimation of the SOH for LIBs can not only improve the safety and reliability of EVs but also has great significance for the development of the EVs industry (Jia et al., 2021).

Many scholars have investigated SOH estimation problems and already proposed a variety of SOH estimation methods, which can be divided into two categories: model-based approaches and data-driven approaches (Akkem et al., 2023). Various degradation models are formulated by analyzing the electrochemical and thermodynamic properties of the LIBs' life cycle (Tian et al., 2021). Doyle et al. (1993) developed an electrochemical model using partial differential equations to characterize the internal electrochemical kinetics. Tsang et al. (Tsang and Chan, 2013) determined the relationship between the equivalent resistance and the capacity of LIBs at different operating temperatures

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for different health conditions, using fuzzy logic for reasoning about the battery SOH. Particularly, the model-based approaches are often combined with filtering algorithms to obtain robustness. For example, Plett et al. (Plett, 2004) used an equivalent circuit model combined with an extended Kalman filter to predict the battery SOH, but using extended Kalman filtering requires complex Jacobi calculations. While Plett et al. (Plett, 2006) introduced a Sigma-Point Kalman filter to implement state estimation. Similarly, Andre et al. (2013) proposed a method using a dual filter to first estimate the capacity of the battery before calculating the SOH. Lai et al. (2021) achieved good estimation accuracy using both particle filtering and extended Kalman filtering. However, the model-based approaches are particularly dependent on the degradation mechanisms of LIBs and require much time to build the correct models. Besides, once any external conditions change, even if a valid model is created, the stability of the model will decrease. Although the combination of filtering models can improve the adaptive capability, the higher computational complexity becomes another issue.

Contrary to the model-based approaches, the data-driven approaches utilize emerging machine learning methods and estimate the battery SOH by mapping features to battery capacity without considering the interpretation of electrochemical principles (Lin et al., 2023), (Lin et al., 2022a). Generally, the data-driven approaches include two parts: one is to extract effective battery degradation features, and the other is to build a suitable machine learning model. During constant current charging, it becomes easier to acquire data on voltage, current, charging time, and temperature. Considering that the environmental impacts on the battery aging process are directly reflected in this data, it provides a strong motivation for the widespread adoption of geometric or statistical methods (Zhang et al., 2024) for analyzing charging curves. However, the traditional battery condition analysis method tends to analyze the rising trend of voltage during the charging process of LIBs, while LIBs will have a voltage plateau during the charging process, so it is difficult to analyze the internal condition of the battery using the voltage curve. Benefiting from the incremental capacity (IC) curves, which show clearly identifiable peaks through the transformation of the voltage plateau phase, these peaks can reflect the loss of active materials within the battery and changes in battery impedance, thus providing a window into the internal condition of the battery. Precisely because IC curves offer in-depth insights into the battery's internal state, their features have been widely applied to reveal SOH (Wu et al., 2021). Therefore, extracting features from IC curves not only aids in a better understanding of the battery's internal mechanisms but also provides crucial data support for accurate prediction of the battery's SOH. Considering the kinetic properties of lithium batteries, the total heat generated during the operation of the battery is primarily composed of Joule heat and reaction heat. The differential temperature (DT) curves reveal how the thermal reactions at the electrodes change during the charging process as the battery ages, thus reflecting the evolution of the battery's internal thermodynamic characteristics (Tian et al., 2020). This analysis highlights the significance of DT curves in understanding the thermodynamic changes during the battery degradation process, providing a method to extract features by analyzing the battery's thermal response to explore the causes of aging.

The primary goal of AI-driven approaches is to build an efficient data-driven model. However, relying solely on single feature data-driven methods fails to comprehensively reveal the process of battery aging from multiple dimensions and does not fully leverage the potential of data-driven approaches. As a result, constructing a comprehensive AI-driven model becomes extremely challenging. Another challenge is that using an unoptimized set of features can lead the model to become stuck in local optima, which further limits improvements in the accuracy of SOH estimation. With the advancement of sensor technology, modern sensors can continuously and real-time monitor multiple complex parameters of the battery. This means that it is now easier and faster to collect a large amount of battery data (Wang et al., 2022). There have been some works (Lin et al., 2022b)–(Liu et al., 2018) that have

addressed the issue of feature set optimization. For example, Lin and Wu et al. (Lin et al., 2022b), (Wu et al., 2023a) developed a multi-feature fusion technique using ensemble learning (EL) to enhance the robustness of SOH predictions. The motivation behind this method lies in integrating different models to process various features, thereby combining the strengths of each model to significantly improve the accuracy of the final estimation model. Lai et al. (2023) used neural networks (NN) to reconstruct the voltage curve and optimize the feature set derived from it. Meanwhile, Hong et al. (2020) also established a deep learning (DL) model that leverages the advantages of automatic feature extraction by deep learning to predict SOH. Furthermore, Liu et al. (2018) adopted PCA to reduce the experimental data and improve the data's correlation. However, the number of potential hidden variables is not considered, and the original data information cannot be retained better. Therefore, it becomes necessary to determine how many new variables should be considered for further study. Although the aforementioned methods can enhance the accuracy of models to a certain extent by increasing the number of features, having more features does not necessarily mean that all of them are beneficial to the model. Adjusting and considering potential hidden variables to improve the accuracy of SOH predictions presents a challenge. Another limitation is that directly adopting heterogeneous methods makes it difficult to fine-tune, and the reliability of statistical features cannot always be guaranteed. Existing methods have not fully utilized multi-source features, nor have they effectively addressed the issue of optimizing feature combinations to eliminate redundancy when using these features.

Thanks to the advances in machine learning methods, many methods have been used for battery SOH estimation. From classical Gaussian regression (Liu et al., 2019), support vector machines (Feng et al., 2019), plain Bayesian (Hu et al., 2015), and neural networks (Wu et al., 2023b), (Mazzi et al., 2024) to deep learning methods such as deep neural networks (Hong et al., 2020), long short-term memory networks (Zhang et al., 2020), (Guo et al., 2024), and fusion of deep learning models to more complex ones. However, relying only on the model does not guarantee the robustness of the prediction results, and the complexity of the model often brings problems such as increased computational size and prediction time. Recently, Chemali et al. (2022) obtained excellent estimability in terms of accuracy and computational efficiency using a convolutional neural network (CNN) model appropriately modified in terms of CNN size and structure. Compared with other deep learning models, CNN can extract more effective information from the data for processing and reduce the number of parameters and prediction time due to its convolutional kernel feature. The performance of CNN not only depends on the size and number of convolution kernels but also has a great relationship with data input dimension and feature salience.

To address the above limitations, a novel SOH estimation method for LIBs is proposed considering feature optimization and CNN. The main contributions of this paper are three-fold.

- 1) Considering the thermodynamic and electrochemical properties, the thermoelectric coupling feature set is extracted from the voltage, IC, and DT curves during the battery aging process.
- 2) To optimally reduce feature redundancy, the feature dimensions are optimized based on a simulated annealing algorithm (SA) combined with cross-validation to improve the prediction performance of the CNN model.
- 3) The optimal features are obtained as the input of the CNN by PCA with the optimal dimensionality, we conduct feature comparison experiments on the Oxford dataset (Birkl, 2017a) to verify the robustness of the proposed method.

The rest of this paper is organized as follows. Section 2 presents the dataset and the principle of feature selection. In section 3, the PCA, CNN, and proposed prediction methods are introduced. Section 4 analyzes the experimental results. Section 5 gives the conclusions.

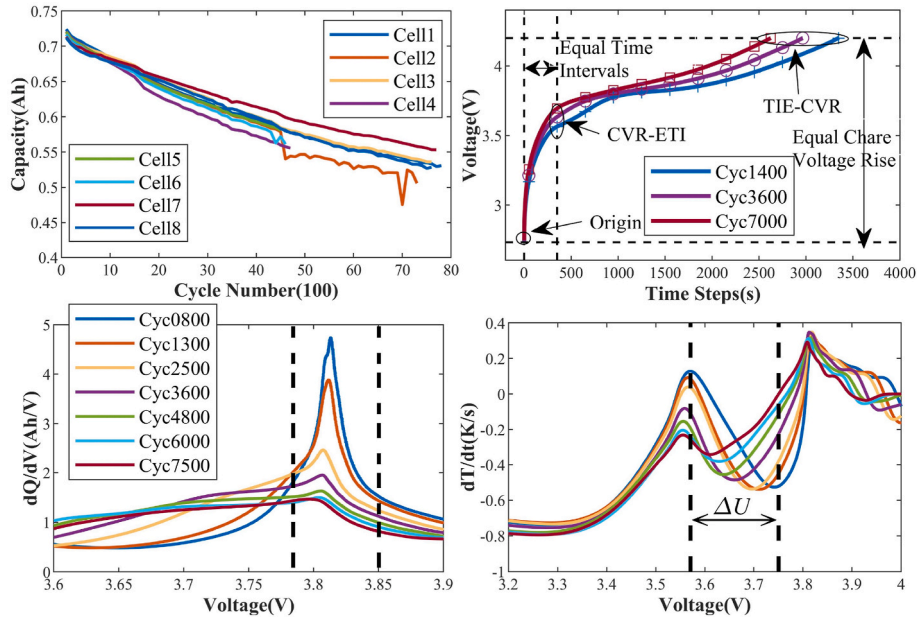


Fig. 1. (a) is the capacity evolution curves. (b), (c) and (d) are the curves of voltage, incremental capacity, and differential temperature in cell 1.

2. Data preparation

2.1. Oxford dataset

The Oxford dataset recorded cycle data for eight Kokam pouch LIBs with 740 mAh. During the measurement, LIBs are labeled as cells 1 to 8, respectively. The positive electrode materials for LIBs are lithium-nickel-cobalt oxide and lithium-cobalt oxide, and the negative electrode material is graphite. To simulate the actual driving conditions of EVs, eight LIBs were charged at 2C and discharged using a dynamic curve during the experiment. After every 100 charging and discharging cycles, the LIBs were charged at a constant current of 1C–4.2V, then switched to constant voltage until the current dropped to the full charge level, recording the time (t), voltage (V), temperature (T), and current (I). This procedure was continued until the battery's SOH decreased to 80%, with the battery temperature fluctuating between 39° and 41°. Fig. 1(a) plots the capacity evolution curves for the eight cells on the Oxford dataset.

2.2. Definition of SOH

The aging of LIBs is manifested by the decay of capacity and the rise of internal resistance. From the perspective of the actual driving condition of EVs, the capacity is more appropriate to calculate the SOH of LIBs, mainly because the capacity is closely related to the range of EVs. The SOH is calculated as shown in Equation (1).

$$SOH = Q_i / Q_l \quad (1)$$

where Q_i denotes the maximum capacity during the i -th cycle and Q_l is the initial rated capacity.

2.3. Voltage and time (Vt) feature extraction

The battery charging and discharging process lead to the accumulation of lithium oxide and electrolyte decomposition, which causes an increase in the voltage rise rate of the battery. As the cycle's number increases, the terminal voltage in the constant-current charging stage rises significantly faster and the constant-current charging time becomes shorter in Fig. 1(b). In fact, under the conditions of the same sampling rate and voltage interval, a new battery requires 1192 sampling points

for the voltage to increase from 3.9V to 4.2V, whereas an aged battery only needs 1163 sampling points. In the experiments, the sampling rate of the battery testing device was 0.1Hz, meaning that during the first cycle, it takes 200 min for the voltage to increase by 0.3V. In contrast, for an aged battery, the same voltage increase can be achieved in only 194 min. Thus, the time required for equal charging voltage rise value and the voltage rise values in the same charge duration are directly extracted from the voltage curves as the aging features (F1~F4). As the cycle's number increases, the time required for the same voltage rise value is shorter, and the voltage rises more for the same charge duration.

2.4. IC feature extraction

The IC curves can be used to analyze the battery aging mechanism at the electrode level. A significant advantage is that the original voltage plateau of the voltage curve is converted into clearly identifiable peaks by IC. The IC peak reflects LIBs' phase change during the insertion and delamination of the Lithium-ion active material. The IC is calculated as follows.

$$IC = dQ/dV \quad (2)$$

where V represents the voltage, and Q represents the capacity.

IC analyzes the battery aging mechanism from the perspective of the electrode layer. Fig. 1(c) shows the IC peaks fall in the interval of 3.78V–3.85V. As the number of cycles increases, the IC curve becomes flatter and the peak becomes lower. The IC curve is very different from the Vt curve, and the peak in the IC curve have a unique shape, height, and position. It can effectively address the voltage plateau issue, serving as a potent indicator for identifying aging mechanisms and online SOH estimation. For this reason, his paper extracts the peak points of IC and the points on the left and right of the same voltage sampling point, totaling 11 features. For example, the battery voltage plateau is translated into clearly identifiable peaks (F5~F7) on the IC curve, and the differentiation is very apparent for the IC curve variances indicative of different aging stages on either side of the 3.82V peak (F8~F15).

2.5. DT feature extraction

LIBs aging process is a thermoelectric coupling process. The rate of change of battery surface temperature often also reflects the degree of

Table 1
Advantages and limitations of the features.

Feature Type	Advantages	Limitations
Vt features	Directly reflects the battery's voltage rise rate and charging time changes due to aging.	May result in a voltage plateau phase.
IC features	Effectively addresses the voltage plateau issue and serves as a potent indicator for aging mechanisms.	IC curve interpretation can be complex due to unique peak shapes and positions.
DT features	Reflects the degree of battery aging through changes in battery surface temperature and avoids the voltage plateau issue.	Temperature measurements can be influenced by external environmental conditions.

Table 2
Feature descriptions.

Feature Type	No.	Feature Description
Vt features	F1	Charge voltage rise of equal time interval(Δt_1)
	F2	Time interval of equal charge voltage rise(Δv_1)
	F3	Charge voltage rise of equal time interval(Δt_2)
	F4	Time interval of equal charge voltage rise(Δv_2)
IC features	F5	Height of IC peak
	F6	Voltage of reaching IC peak
	F7	Time(t) of reaching IC peak
	F8	Height of t+1
	F9	Voltage of t+1
	F10	Height of t-1
	F11	Voltage of t-1
	F12	Height of t+2
	F13	Voltage of t+2
	F14	Height of t-2
	F15	Voltage of t-2
DT features	F16	Height(H_{peak}) of DT peak
	F17	Height(H_{valley}) of DT valley
	F18	Voltage(V_{peak}) of reaching DT peak
	F19	Voltage(V_{valley}) of reaching DT valley
	F20	Time(t_{peak}) of reaching DT peak
	F21	Time(t_{valley}) of reaching DT valley
	F22	Difference between (H_{peak}) and Voltage Height(H_{valley})
	F23	Difference between Voltage(V_{peak}) and Voltage(V_{valley})
	F24	Difference between Time(t_{peak}) and Time(t_{valley})

battery aging. Direct calculation using discrete data is sensitive to temperature measurement noise. To solve this issue, the rate of temperature change is approximated by finite difference method. DT is calculated as follows,

$$DT(k) = dT/dt \quad (3)$$

where T is the temperature, and t represents the time.

Temperature is also a crucial source of information in BMS systems.

From an engineering perspective, the rate of change in battery surface temperature often reflects the degree of battery aging. A potential reason for the variation in the DT curve is the loss of active material in the cathode, which has been found to be more pronounced than in the anode in the Oxford dataset (Birkel, 2017b). From Fig. 1(d), it can be observed that the DT curve also effectively avoids the voltage plateau issue. Unlike the IC curve, the DT curve exhibits not only peaks but also troughs. The DT curve reaches the first peak point between 3.5V and 3.6V. Then drops to a valley value point between charging 3.6V and 3.8V. As the number of cycles increases, the first peak point moves down to the left and the valley value point moves up to the left. The first peak point, the valley value point, and the interval between them are used as aging features (F16 , F17 , F22~24). We also sampled the voltage and time of the first peak and valley points as aging features (F18~21).

To comprehensively describe the aging phenomenon of LIBs, this article extracted 24 features from voltage, IC, and DT curves, reflecting the battery aging dynamics from various perspectives. All the feature descriptions, dvantages and limitations are shown in Table 1 and

Table 3
Parameters of CNN.

	Layers	Parameters
CNN block 1	1D convolution layer	$k : 64$, kernel size : 2, stride : 1, $d : 1$
	Activation function	ReLU
CNN block 2	1D convolution layer	$k : 16$, kernel size : 2, stride : 1, $d : 1$
	Activation function	ReLU
Pooling block	1D max pooling layer	kernel size : 2, stride : 2
Flatten block	flatten layer	
FC block 1	full connection layer	$k : 1024$, drop out : 0.5
	Activation function	ReLU
FC block 2	full connection layer	$k : 512$, drop out : 0.5
	Activation function	ReLU
FC block 3	full connection layer	$k : 1$
	Activation function	Sigmoid

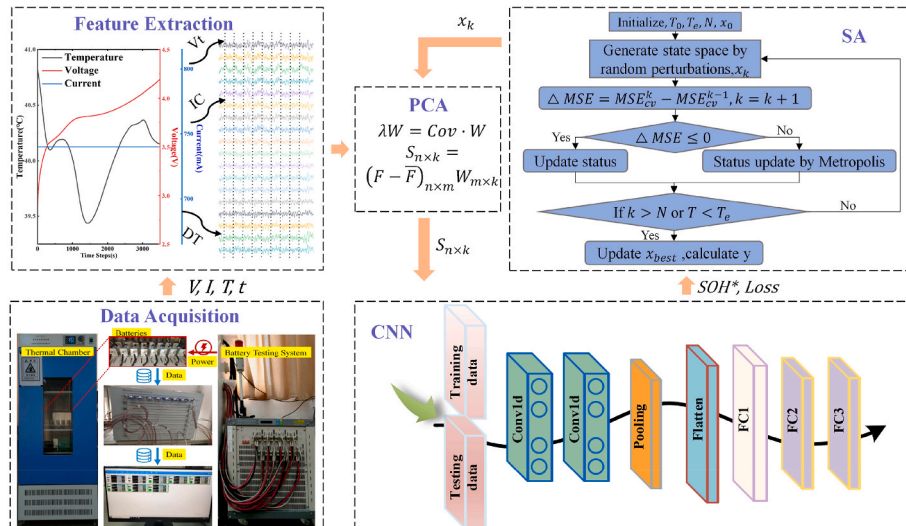


Fig. 2. The flow of the proposed SOH estimation method.

Table 4
Algorithm flow.

Definition:

$$\begin{cases} \Delta MSE = MSE_{cv}^z - MSE_{cv}^{z-1} \\ P = e^{-|f(k_z) - f(k_{z-1})| \times C_T} \end{cases}$$

where k_z is the dimension of the features, C_T presents the control coefficient.

The number of update iterations $z = z + 1$.

Initialization parameters:

Set the initial control parameter $T_0=100$, the value $T_{end}=0$ when the iteration stops, the maximum number of iterations $N=15$, the generated initial value $k_0=12$ and calculate MSE_{cv}^0 .

Step 1:

Randomly generated new solution k_z .

Step 2:

Calculate the corresponding objective function value MSE_{cv}^z .

If $\Delta MSE \leq 0$ & $\Delta \mathcal{R} \leq 0$, update the current optimal solution $k_{best} = k_z$.

If $\Delta MSE > 0$, according to the Metropolis criterion, enter the next state with probability P.

Step3:

Repeat steps 1 and 2 until the number of iterations reaches $N=15$ or the control parameter is reduced to T_{end} . Update the optimal state value and calculate the SOH value.

Table 5
Values of k_0 , and k_z .

No.	1(k_0)	2	3	4	5	6	7	8	9	10	11	12	13	14	15
k_z	12	16	14	13	18	10	11	16	5	6	9	18	8	3	10

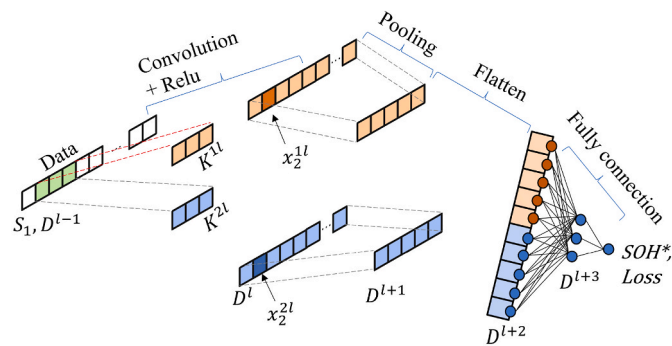


Fig. 3. 1-dimensional CNN structure.

Table 2.

3. Methodologies of the proposed method

The procedure of the proposed SOH estimation method is shown in Fig. 2. Firstly, the raw data of voltage, current, temperature, and time are obtained in the aging cycle experiment. The aging features of the full factor are extracted based on the mechanism analysis and data processing means. Secondly, the CNN estimator is constructed. The parameters of CNN are shown in Table 3. Subsequently, the feature dimensions are optimized through the collaborative invocation of SA and PCA. Here, PCA is employed to reduce the feature dimensions, while SA is utilized to determine the optimal dimensionality of the features post-PCA optimization based on MSE. Subsequently, the best features, which are the principal components (PCs), are inputted into the CNN to achieve precise SOH estimation. The dimension values randomly generated during the collaborative process of SA and PCA are detailed in Table 4.

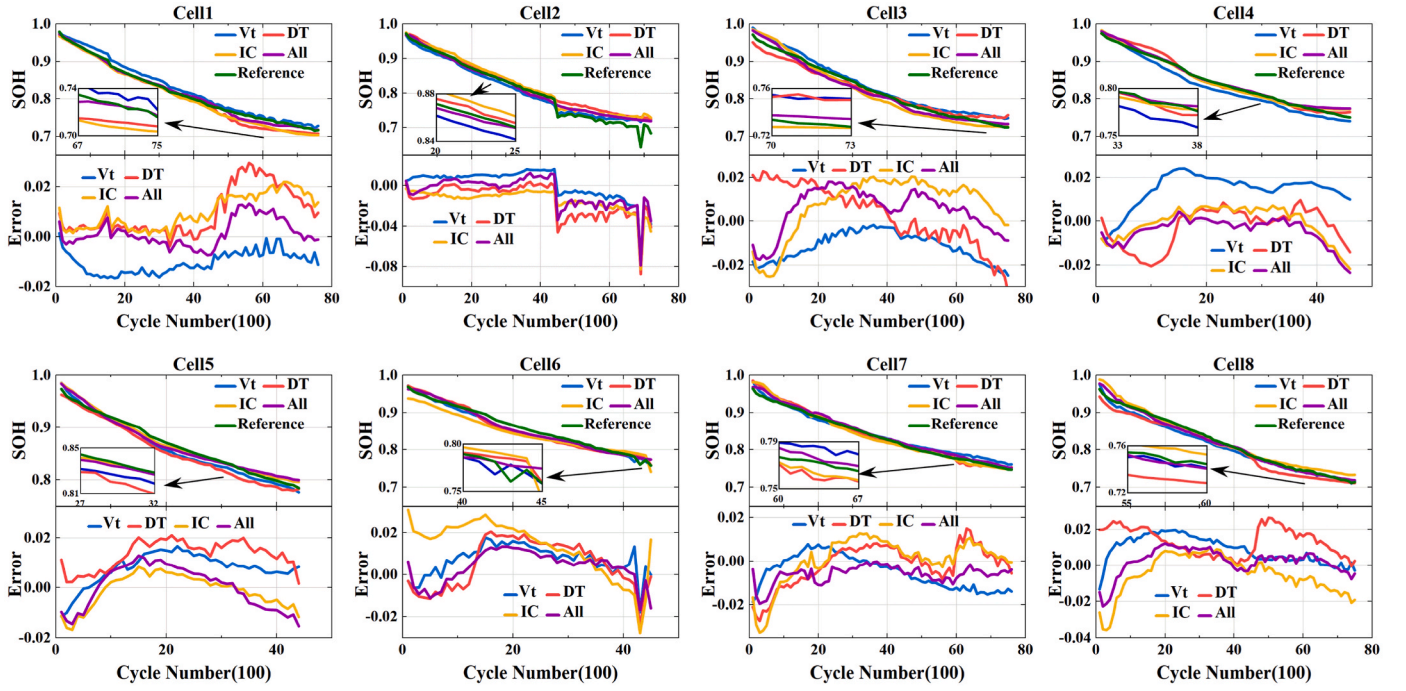


Fig. 4. The experimental results of different features on the Oxford dataset.

Table 6

The mean GRA of different features.

GRA	Vt	DT	IC
Average	72.22%	80.99%	95.12%

3.1. Feature dimensionality reduction

PCA is a statistical method of dimensionality reduction. Compared with other feature optimization algorithms, it not only can effectively

reduce the data dimension but also can retain the important information in the original data. The dimensionality of the feature description is often increased to comprehensively describe the aging phenomenon of LIBs. However, as the number of features increases, the computational complexity increases. There is redundant information between the features. The accuracy does not necessarily increase. To ensure both the battery aging process description and reduce the redundancy of data and computation, this paper uses PCA to reduce the dimensionality of the features. The specific steps are as follows.

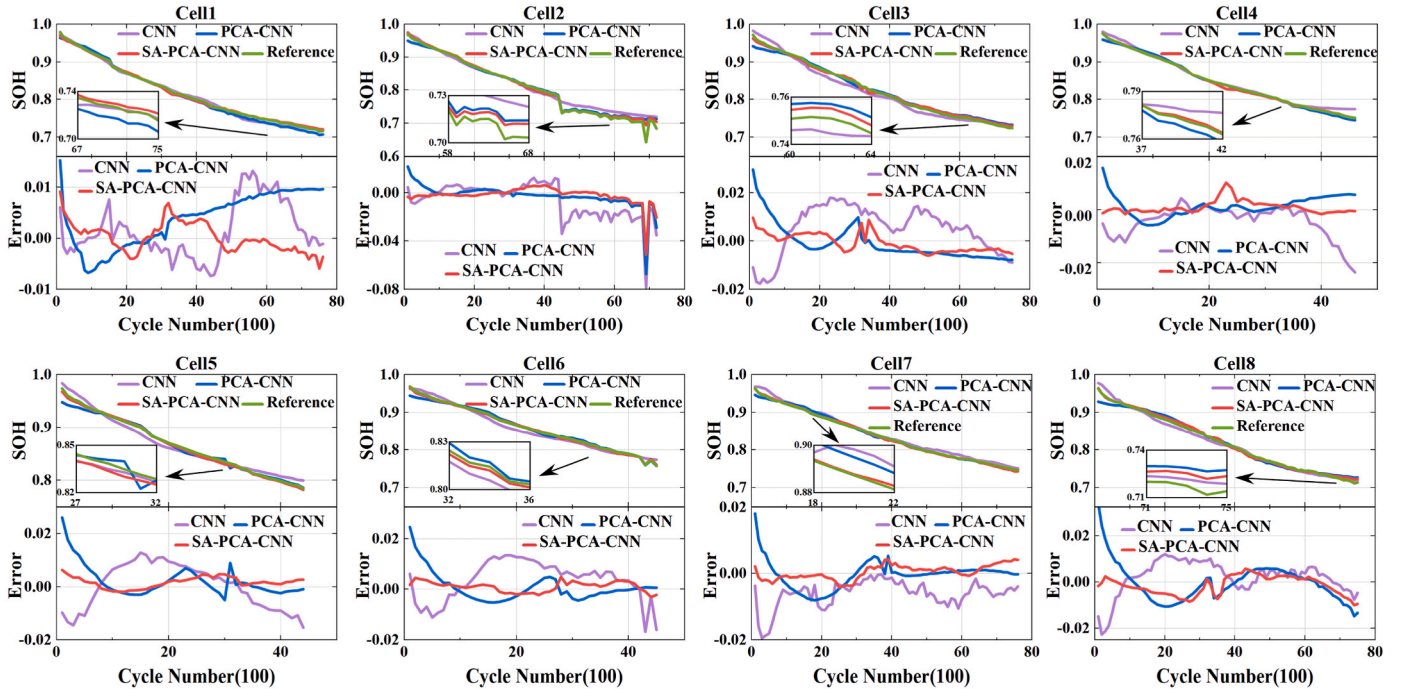


Fig. 5. The experimental results of different models on the Oxford dataset.

Table 7
SOH estimation errors of different features on the Oxford dataset.

No.	Methods	MAE	RMSE	MAPE	R ²
Cell1	Vt	1.03%	1.11%	1.27%	97.62
	DT	0.98%	1.34%	1.28%	96.59
	IC	0.98%	1.19%	1.27%	97.33
	All	0.41%	0.56%	0.53%	99.42
Cell2	Vt	1.22%	1.50%	1.59%	96.73
	DT	1.50%	2.18%	2.06%	93.18
	IC	1.46%	1.85%	1.95%	95.06
	All	1.13%	1.61%	1.55%	96.26
Cell3	Vt	1.17%	1.34%	1.42%	96.51
	DT	1.18%	1.38%	1.41%	96.31
	IC	1.37%	1.50%	1.67%	95.54
	All	0.97%	1.10%	1.16%	97.64
Cell4	Vt	1.51%	1.61%	1.83%	93.44
	DT	0.71%	0.92%	0.83%	97.83
	IC	0.60%	0.75%	0.73%	98.60
	All	0.51%	0.76%	0.62%	98.63
Cell5	Vt	0.91%	1.00%	1.06%	96.24
	DT	1.32%	1.44%	1.56%	92.09
	IC	0.55%	0.67%	0.61%	98.36
	All	0.77%	0.85%	0.88%	97.28
Cell6	Vt	0.86%	0.99%	1.02%	96.75
	DT	1.08%	1.24%	1.28%	94.97
	IC	1.65%	1.83%	1.87%	89.73
	All	0.75%	0.86%	0.88%	97.61
Cell7	Vt	0.73%	0.90%	0.89%	97.92
	DT	0.74%	0.95%	0.84%	97.67
	IC	0.75%	1.04%	0.85%	97.10
	All	0.61%	0.73%	0.72%	98.53
Cell8	Vt	0.94%	1.13%	1.10%	97.47
	DT	1.32%	1.53%	1.59%	95.42
	IC	0.91%	1.19%	1.09%	97.34
	All	0.60%	0.76%	0.70%	98.90
Average	Vt	1.05%	1.20%	1.27%	96.59
	DT	1.10%	1.37%	1.36%	95.51
	IC	1.03%	1.25%	1.26%	96.13
	All	0.72%	0.90%	0.88%	98.04

Table 8
SOH estimation errors of different models on the Oxford dataset.

No.	Models	MAE	RMSE	MAPE	R ²
Cell1	CNN	0.41%	0.56%	0.53%	99.42
	PCA-CNN	0.55%	0.64%	0.69%	99.27
	SA-PCA-CNN	0.24%	0.29%	0.28%	99.86
Cell2	CNN	1.13%	1.61%	1.55%	96.26
	PCA-CNN	0.54%	1.03%	0.69%	98.57
	SA-PCA-CNN	0.29%	0.51%	0.51%	99.20
Cell3	CNN	0.97%	1.10%	1.16%	97.64
	PCA-CNN	0.57%	0.74%	0.65%	99.14
	SA-PCA-CNN	0.30%	0.33%	0.42%	99.73
Cell4	CNN	0.51%	0.76%	0.62%	98.63
	PCA-CNN	0.31%	0.42%	0.34%	99.68
	SA-PCA-CNN	0.16%	0.26%	0.20%	99.83
Cell5	CNN	0.77%	0.85%	0.88%	97.28
	PCA-CNN	0.40%	0.64%	0.39%	99.02
	SA-PCA-CNN	0.26%	0.26%	0.25%	99.77
Cell6	CNN	0.75%	0.86%	0.88%	97.61
	PCA-CNN	0.38%	0.60%	0.38%	99.29
	SA-PCA-CNN	0.20%	0.22%	0.23%	99.84
Cell7	CNN	0.61%	0.73%	0.72%	98.53
	PCA-CNN	0.28%	0.43%	0.29%	99.62
	SA-PCA-CNN	0.19%	0.21%	0.22%	99.87
Cell8	CNN	0.60%	0.76%	0.70%	98.03
	PCA-CNN	0.62%	0.83%	0.71%	98.93
	SA-PCA-CNN	0.31%	0.37%	0.45%	99.62
Average	CNN	0.72%	0.90%	0.88%	95.54
	PCA-CNN	0.46%	0.67%	0.52%	99.19
	SA-PCA-CNN	0.24%	0.31%	0.32%	99.72

Step 1: Calculate the mean value. $F = [f^1, f^2, \dots, f^n]^T$ denotes the battery aging features, where n is the number of cycles. Each sample $x^i = [f_1^i, f_2^i, \dots, f_m^i]$ has m dimensions. Calculate the mean value $\bar{F}$ of F .

Step 2: Calculate the covariance matrix.

$$\text{Cov}(F, F) = E[(F - \bar{F})(F - \bar{F})^T] \quad (4)$$

Step 3: Solve for the eigenvalues and corresponding eigenvectors of the covariance matrix.

$$\lambda W = \text{Cov} \cdot W \quad (5)$$

where $\lambda = [\lambda_1, \lambda_2, \dots, \lambda_m]^T$ is the eigenvalue vector and W is the corresponding eigenvector matrix. The eigenvalues are sorted from largest to smallest. The largest k eigenvalues are selected.

Step 4: Calculate the reduced dimensional feature matrix. The variation in the values of k is shown in Table 5.

$$S_{n \times k} = (F - \bar{F})_{n \times m} W_{m \times k} \quad (6)$$

3.2. CNN estimation

CNN reinforces the relevance of the input data in the convolution layer to effectively extract features and reduces the dimensionality by sharing weights and pooling layers, making it smaller in scale compared to a typical neural network. The input data of CNN is aging features set $D = \{(S_1, SOH_1), (S_2, SOH_2), \dots, (S_i, SOH_i), \dots, (S_n, SOH_n)\}^T$ after dimensionality reduction by PCA.

According to the characteristics of the input data, this paper uses 1-dimensional CNN to estimate the SOH of LIBs. Fig. 3 shows the structure of 1-dimensional CNN. The feature map D^l is obtained from the previous layer feature map D^{l-1} after filtering by a convolution kernel. For the convolutional output x_i^{ml} can be described as the result of filtering the data of x_i^{ml} by the m -th convolution kernel in the convolutional layer of the l -th layer, and its filtering operation can be expressed by the following equation.

$$x_i^{ml} = \sigma_{cnn} \left(\sum_{i \in M_j^{l-1}} x_i^{kl-1} K_i^{ml} + b_j^{ml} \right) \quad (7)$$

where σ_{cnn} denotes the activation function used in the convolution layer, K_i^{ml} is the i -th weight coefficient of the m -th convolution kernel in the convolution layer of the l -th layer, b_j^{ml} is the bias parameter.

CNN convolutional filtering requires downsampling to reduce the data size.

$$D^{l+1} = \text{down}_{cnn}(D^l) \quad (8)$$

where $\text{down}_{cnn}(\cdot)$ represents the downsampling function. Then comes the fully connected layer, whose operation process is simpler compared to the convolutional operation, as in the following equation.

$$SOH^* = f(D^{l+2}) = \sigma_{cnn} \left(\sum_{i \in M_j^{l+2}} x_i^{l+2} K_i^{l+3} + b_j^{l+3} \right) \quad (9)$$

In the above equation, D^{l+2} is the result after tiling the feature map D^{l+1} in the pooling layer, b_j^{l+3} and K_i^{l+3} are the i -th bias constant and weight parameters in the fully connected layer.

The specific CNN structure used in the algorithm proposed in this article is illustrated in Table 3. In CNN block1, CNN block2, and the Pooling block, a smaller kernel size (2x1) has been selected. This

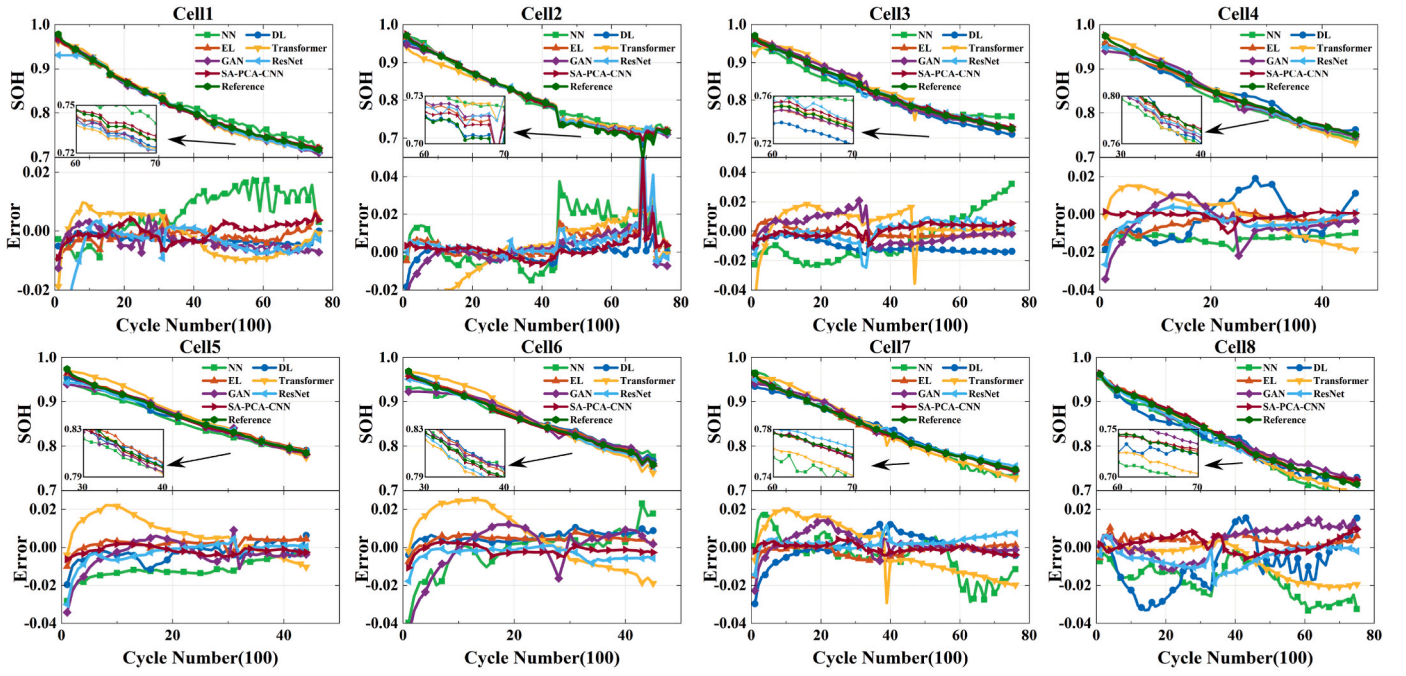


Fig. 6. SOH estimation of different methods.

approach reduces the number of model parameters while capturing closely related feature variations. The stride is set to 1, aimed at minimizing the reduction in input feature size, which helps preserve more detail information. To extract as many features as possible, the number of convolutional kernels, k , is set to 64 in CNN block1 and 16 in CNN block2. Regarding activation functions, ReLU is utilized in most layers to prevent the issue of gradient vanishing and to enhance computation efficiency. Additionally, the Sigmoid activation function is adopted in the output layer, suitable for constraining the output results within the range of 0–1, which fits the SOH regression prediction requirements.

3.3. Feature optimization

The full features are selected as the training data. The PCA is adopted to reduce the dimensionality of features for retaining the important features while reducing the computational effort of the CNN estimation model. While offering superior interpretability and efficiency, its actual performance is not far behind other dimensionality reduction methods. Getting the best dimensionality is the key for CNN to improve the estimation performance. SA has higher convergence speed, better optimization performance, and is easy to implement and understand. This paper uses SA to find the optimal feature dimension, saving computational resources and improving estimation accuracy.

During the search process of SA, cross-validation is applied to evaluate the CNN with the new feature dimension of input. the average mean square deviation of cross-validation is set as the internal energy as follows.

$$MSE = \sqrt{Loss} = \sqrt{\frac{1}{n} \sum_{i=1}^n (SOH_i^* - SOH_i)^2} \quad (10)$$

where $Loss$ is the loss function of CNN.

The steps of the SA-based feature optimization method (SA-PCA-CNN) are shown in Table 4. In the process of optimizing the feature combination to reduce redundancy while simultaneously enhancing the prediction accuracy of battery SOH, the selection and determination of the optimal features primarily adhere to two criteria: first, it is essential to ensure that the overall redundancy of the input variables is reduced; second, the selected features must enhance the accuracy of the battery

SOH prediction. As illustrated in Step 2, a new solution is accepted only if it not only decreases the objective function value ($\Delta MSE \leq 0$) but also improves or maintains the related optimization metrics ($\Delta \mathcal{R} \mathcal{R} \mathcal{R}$). This standard ensures that the method continually evolves towards a better solution throughout the process of seeking the optimal feature set.

4. Experiments and analysis

To verify the performance of the proposed method, feature validity and model validity comparison experiments are carried out on the Oxford dataset. The results and errors can be seen in the degree of deviation between the estimation results and the actual results on a macro level. To evaluate the experimental results scientifically, two metrics are applied, namely, the mean absolute error (MAE), the root mean square error (RMSE), the mean absolute percentage error (MAPE), and the coefficient of determination (R^2). In addition, in mathematical terms, we evaluate feature redundancy $\mathcal{R}$ by calculating the Spearman correlation coefficient across feature channels.

$$\left\{ \begin{array}{l} MAE = \frac{1}{m} \sum_{i=1}^m |SOH_i^* - SOH_i| \\ RMSE = \sqrt{\frac{1}{m} \sum_{i=1}^m (SOH_i^* - SOH_i)^2} \\ MAPE = \frac{1}{n} \sum_{i=1}^m \left| \frac{SOH_i - SOH_i^*}{SOH_i} \right| \\ R^2 = 1 - \frac{\sum_{i=1}^n (SOH_i - SOH_i^*)^2}{\sum_{i=1}^n (SOH_i - \bar{SOH})^2} \end{array} \right. \quad (12)$$

$$\left\{ \begin{array}{l} \mathcal{R} = \frac{1}{c(c-1)} \sum_i \sum_{j \neq i} |\rho(\mathbf{F}_i, \mathbf{F}_j)| \\ \rho(\mathbf{F}_i, \mathbf{F}_j) = 1 - \frac{6 \sum d^2(\mathbf{F}_i, \mathbf{F}_j)}{m(m^2 - 1)} \end{array} \right. \quad (13)$$

Table 9
SOH estimation errors of different methods.

Cell	Methods	MAE	RMSE	MAPE	R ²
Cell1	NN	0.82%	0.97%	1.06%	98.21
	DL	0.29%	0.32%	0.36%	99.81
	EL	0.24%	0.30%	0.27%	99.85
	Transformer	0.60%	0.68%	0.73%	99.20
	GAN	0.41%	0.47%	0.51%	99.63
	ResNet	0.49%	0.78%	0.59%	98.87
Cell2	SA-PCA-CNN	0.24%	0.29%	0.28%	99.86
	NN	1.36%	1.81%	1.86%	95.22
	DL	0.31%	0.55%	0.38%	99.62
	EL	0.28%	0.94%	0.81%	98.72
	Transformer	1.41%	1.79%	1.79%	95.44
	GAN	0.61%	0.95%	0.82%	98.72
Cell3	ResNet	0.61%	1.18%	0.84%	98.01
	SA-PCA-CNN	0.38%	0.75%	0.81%	99.20
	NN	1.47%	1.64%	1.80%	94.67
	DL	1.03%	1.11%	1.31%	97.53
	EL	0.28%	0.34%	0.35%	99.77
	Transformer	0.81%	1.08%	0.94%	97.71
Cell4	GAN	0.72%	0.84%	0.86%	98.61
	ResNet	0.49%	0.64%	0.61%	99.19
	SA-PCA-CNN	0.34%	0.39%	0.41%	99.73
	NN	1.27%	1.00%	1.50%	95.94
	DL	0.95%	0.83%	1.10%	97.26
	EL	0.46%	0.44%	0.50%	99.31
Cell5	Transformer	0.96%	1.09%	1.15%	97.00
	GAN	0.76%	0.92%	0.90%	97.90
	ResNet	0.38%	0.47%	0.45%	99.45
	SA-PCA-CNN	0.16%	0.26%	0.20%	99.83
	NN	1.17%	1.27%	1.28%	94.51
	DL	0.44%	0.57%	0.46%	99.09
Cell6	EL	0.29%	0.34%	0.33%	99.66
	Transformer	0.97%	1.15%	1.09%	95.01
	GAN	0.56%	0.70%	0.63%	98.13
	ResNet	0.30%	0.45%	0.33%	99.23
	SA-PCA-CNN	0.26%	0.26%	0.25%	99.77
	NN	0.77%	1.13%	0.81%	97.01
Cell7	DL	0.57%	0.61%	0.68%	98.88
	EL	0.48%	0.50%	0.55%	99.21
	Transformer	1.39%	1.58%	1.61%	92.02
	GAN	0.85%	1.08%	0.98%	96.26
	ResNet	0.32%	0.40%	0.38%	99.49
	SA-PCA-CNN	0.20%	0.22%	0.25%	99.82
Cell8	NN	0.82%	1.12%	1.02%	96.52
	DL	0.50%	0.70%	0.56%	98.95
	EL	0.23%	0.35%	0.25%	99.74
	Transformer	1.27%	1.37%	1.53%	94.87
	GAN	0.38%	0.56%	0.44%	99.15
	ResNet	0.30%	0.38%	0.37%	99.61
Average	SA-PCA-CNN	0.19%	0.21%	0.22%	99.87
	NN	1.73%	1.95%	2.24%	92.26
	DL	1.34%	1.61%	1.64%	94.71
	EL	0.38%	0.43%	0.48%	99.63
	Transformer	0.79%	1.13%	1.05%	97.45
	GAN	0.79%	0.87%	1.00%	98.50
Average	ResNet	0.57%	0.73%	0.68%	98.94
	SA-PCA-CNN	0.36%	0.43%	0.45%	99.63
	NN	1.18%	1.36%	1.45%	95.54
	DL	0.68%	0.79%	0.81%	98.23
	EL	0.33%	0.46%	0.44%	99.45
	Transformer	1.03%	1.23%	1.24%	96.09
Average	GAN	0.64%	0.80%	0.77%	98.36
	ResNet	0.43%	0.63%	0.53%	99.10
	SA-PCA-CNN	0.27%	0.35%	0.36%	99.71

where c represents the feature dimension, and $\rho(F_i, F_j)$ ected features and produces a value between 0 and 1. Therefore, a higher $\mathcal{R}$ value indicates high redundancy in the dataset, while a lower $\mathcal{R}$ value suggests a better feature subset with less redundancy.

4.1. Feature validity comparison

To verify the performance of the proposed method, feature validity and model validity comparison experiments are carried out on the Oxford dataset. The results and errors can be seen in the degree of deviation between the estimation results and the actual results on a macro level.

The experimental results and errors based on the CNN model with different features are given in Fig. 4. From the comparison of the estimated SOH curve based on single features, The estimated SOH curve based on the full feature is closer to the true SOH value. the different features' errors are displayed in Table 7. The average MAE values of eight cells on Vt, IC, and DT features are 1.05%, 1.1%, and 1.03%, respectively. The average RMSE values of eight cells on Vt, IC, and DT features are 1.20%, 1.37%, and 1.25% respectively. All features' MAE, RMSE, MAPE, and R² are 0.72%, 0.90%, 88%, and 98.04%, respectively, smaller than single features. All features' MAE and RMSE are minimum in cells 1–4 and 6–8 and are also close to the minimum in cell 7. Vt features can directly reflect the electrical properties of LIBs. DT features can reflect the thermodynamic properties of LIBs. IC features can reflect the electrochemical properties of LIBs. The full features describe the LIBs' aging process from multiple perspectives. Therefore full features can better describe the LIBs' aging process. As shown in Table 6, the correlation of the three types of features with SOH also demonstrates their utility in SOH estimation.

4.2. Model validity comparison

To verify the model's effectiveness, we conducted comparison experiments with the standard CNN and PCA-CNN. Fig. 5 displays prediction results and errors for the CNN, PCA-CNN, and SA-PCA-CNN models. As shown in Fig. 5, SA-PCA-CNN's estimated curve is closest to the true value and can track the true value curve relatively smoothly. In particular, two SOH plunges occurred in cell 2. When the first plunge occurs, both SA-PCA-CNN and PC-CNN can track the true value relatively accurately, while CNN shows large fluctuations. All three models fluctuate when there is a second plunge, but the SA-PCA-CNN has the smallest estimation error. The synergistic effect of SA and PCA largely filters out the insignificant features, reduces the influence of noise data, and retains the favorable features for SOH estimation.

The corresponding error values are given in Table 8. The SA-PCA-CNN achieves the minimum MAE and RMSE from cells 1–8. The average MAE and RMSE of SA-PCA-CNN are 0.27% and 0.35% respectively, both less than 0.50%. Compared with CNN, the average MAE and RMSE of PCA-CNN reduced by 0.71% and 0.90% respectively. Compared with PCA-CNN, the average MAE and RMSE of SA-PCA-CNN reduced by 0.46% and 0.67%, respectively. In cell 2, where two SOH plunges occurred, the SA-PCA-CNN estimation errors are significantly reduced relative to the other two models. PCA-CNN outperforms CNN in terms of estimation results because extracting full features will cause redundancy in some of the data, and PCA dimensionality reduction removes the redundant parts from the initial data and better describes the battery aging process. However, fixed-dimension PCA relies on manual settings, which may not be optimal. Therefore SA-based feature dimensional optimization algorithm can obtain the best estimation

Table 10
Computational time of different methods.

Method	NN	DL	EL	Transformer	GAN	ResNet	SA-PCA-CNN
Time	13.06s	122.47s	203.81s	116.45s	87.07s	32.63s	71.52s

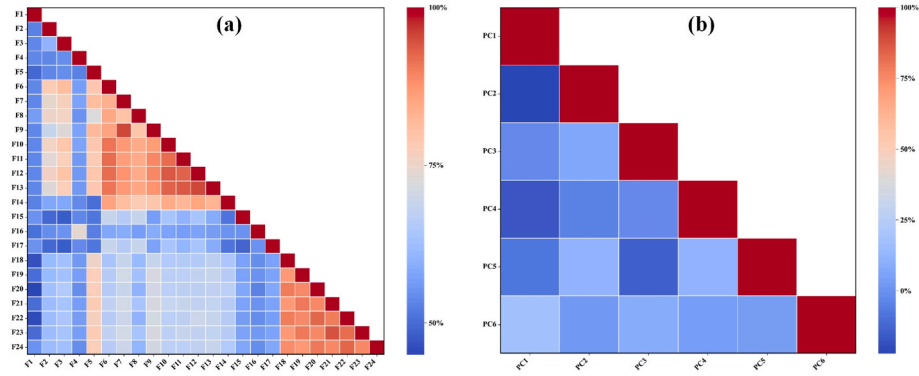


Fig. 7. (a) Spearman correlation (ρ) Heatmap of origin, (b) Spearman correlation (ρ) Heatmap of PCA

Table 11

Values of Redundancy index.

SETS	c	$\mathcal{R}$
sets of origin	24	0.43
sets of PCA	6	0.09 ↓

Table 12

Battery parameters and experimental working conditions.

Number	Nominal Capacity	Nominal Voltage	Charge Rate	Discharge Rate
Cell1	1.5Ah	3.60V	1.5A(1C)	7.5A(5C)
Cell2	1.5Ah	3.60V	1.5A(1C)	15A(10C)
Cell3	1.5Ah	3.60V	3A(2C)	7.5A(5C)
Cell4	1.5Ah	3.60V	3A(2C)	15A(10C)

results. To further validate the efficiency of our proposed method, we assessed the computational burden of training and predicting with our method. The algorithm was tested on a computer with a CPU Intel Core i7-7700 2.80 GHz and 8G RAM. The training and prediction times of our method depend on the base learners' training and prediction times. The training time for the method proposed in this article is not long, only about 72 s, which is entirely sufficient for model updates when the BMS is in a static moment. Moreover, the prediction computational time is around 0.001 s, which is also acceptable.

To demonstrate the advancement of your approach, a comparison with other advanced machine learning methods has been added. The experimental outcomes, as illustrated in Fig. 6 and Table 9, indicate that while the NN network structure is simple and easy to implement, its limited generalization capability leads to inaccuracies when predicting

unseen data. Notably, the GAN achieved an MAE, RMSE, and MAPE of 0.64%, 0.80%, and 0.77% respectively, showing more precise predictive accuracy than the Transformer. This phenomenon can be partially attributed to the GAN's generator, which can produce a subset of effective data that augments the training set, thereby enhancing the model's ability to learn the true characteristics of battery performance degradation. Furthermore, due to ResNet's introduction of residual learning, which prevents the common problem of vanishing gradients in traditional deep networks, the model is capable of learning more complex feature representations. ResNet obtained scores of 0.43%, 0.63%, and 0.53% for MAE, RMSE, and MAPE respectively, and reached 99.10% for R^2 . The experimental results suggest that there is a certain improvement in model accuracy when the data is further optimized in terms of features. However, it is still evident that unoptimized models are slightly inferior to those that are optimized. Among them, the average accuracy of ensemble learning is only second to PCA-CNN-SA. For example, in the experimental results of Cell3, the MAE and RMSE values are 0.29% and 0.26%, respectively, outperforming the PCA-CNN-SA. This is because ensemble learning utilizes multiple machine learning models, which, of course, also increases the complexity of the algorithm.

The runtime of different algorithms is given in Table 10. In terms of computational time, the SA-PCA-CNN method ranks third with a processing time of 71.52 s, outperforming the DL (122.47 s) and Transformer (116.45 s) methods that require numerous parameters, and the EL (203.81 s) that involves multiple models and demands substantial computational power even when run in parallel. It also surpasses the GAN method (87.07 s), which needs to train both a generator and a discriminator. However, the SA-PCA-CNN method is slightly slower than the NN method, which achieves faster computation speed (13.06 s) due to its simpler structure, and is also marginally outpaced by the ResNet (32.63 s).

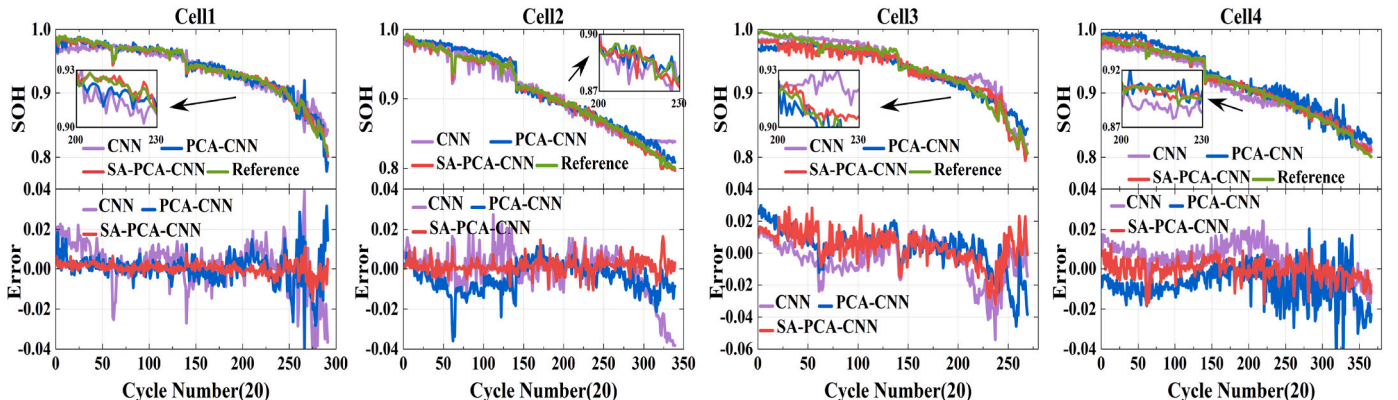


Fig. 8. The experimental results of different methods on the our laboratory dataset.

Table 13

SOH estimation errors of different methods on the our laboratory dataset.

No.	Methods	MAE	RMSE	MAPE	R ²
Cell1	CNN	0.92%	1.23%	1.00%	91.34
	PCA-CNN	0.52%	0.78%	0.57%	96.54
	SA-PCA-CNN	0.21%	0.30%	0.23%	99.49
Cell2	CNN	0.79%	1.01%	0.88%	96.47
	PCA-CNN	0.67%	0.85%	0.74%	97.59
	SA-PCA-CNN	0.28%	0.40%	0.31%	99.48
Cell3	CNN	0.96%	1.28%	1.04%	92.75
	PCA-CNN	1.03%	1.31%	1.11%	92.32
	SA-PCA-CNN	0.91%	1.14%	0.97%	94.26
Cell4	CNN	0.81%	0.94%	0.89%	96.80
	PCA-CNN	0.95%	1.20%	1.06%	94.85
	SA-PCA-CNN	0.36%	0.49%	0.40%	99.15
Average	CNN	0.87%	1.12%	0.95%	94.34
	PCA-CNN	0.79%	1.04%	0.87%	95.32
	SA-PCA-CNN	0.44%	0.58%	0.48%	98.01

4.3. The redundancy index before and after SA-PCA-CNN validity comparison

Based on the definition of feature redundancy presented, we used Cell1 as a case study to compute and evaluate the redundancy index both before and after the implementation of SA-PCA-CNN. The results are illustrated in Fig. 7. From the visuals presented in Fig. 7 (a), we can distinctly discern that the correlation coefficients among the features remain substantially elevated. A significant majority of the pairwise feature relationships showcase a ρ value that sails beyond the 50% mark, and intriguingly, a select few even breach the 90% threshold. Such observations underscore that the features, untouched by SA-PCA-CNN intervention, are enmeshed in a web of pronounced redundancy. In stark contrast, Fig. 6 (b) paints a different picture. After undergoing meticulous optimization via SA-PCA-CNN, the ρ values characterizing pairwise feature correlations have not only been tamed below the 50% boundary, but the correlation coefficient between the first principal component and its fellow counterparts is tantalizingly close to the zero mark. This transformation stands as a robust testament to the prowess of our proposed method in curtailing redundancy with remarkable efficacy.

Considering the analysis of images and data in section 4.2, the SA-PCA-CNN approach is a testament to our commitment to balancing the dual imperatives of accuracy and efficiency, harnessing the best of both worlds to achieve a gold-standard in predictive outcomes.

In Table 11., we further demonstrate that after undergoing PCA processing, the redundancy of the features is significantly reduced. This optimization in reducing redundancy lays a solid foundation for enhancing the accuracy of SOH estimations.

5. Verification of other dataset

To validate the robustness and generalization ability of the method proposed in this paper, additional experimental verifications under different operational condition datasets were conducted. The specific experimental conditions are outlined in Table 12.

As illustrated in Fig. 8 and Table 13, our method demonstrated outstanding predictive accuracy, achieving the lowest values for average MAE, RMSE, and MAPE across the four batteries, recorded at 0.44%, 0.58%, and 0.48%, respectively. Notably, even in scenarios where the data set exhibited frequent battery capacity recovery phenomena, our proposed SA-PCA-CNN algorithm maintained high sensitivity and robust prediction. Compared to the PCA-CNN algorithm without SA optimization, the R² improved by 2.69. These results further validate that our algorithm is not only effective in addressing performance degradation under standard conditions but can also adapt and accurately predict the aging process of batteries under more stringent conditions.

6. Discussions and limitations

This research method failed to consider users' actual charging habits during the modeling process, which could be influenced by various factors such as daily travel patterns, the availability of charging infrastructure, fluctuations in electricity prices, and personal preferences, directly affecting the battery's lifespan and efficiency. This oversight might lead to suboptimal system optimization efficiency or inaccurate battery health assessments in practical BMS applications. Particularly, optimizing the system based on charging cycles that do not match user habits, such as optimizing for long cycles for users who prefer short charging periods, may not effectively extend battery life. Another challenge is that privacy policies and data protection requirements make it difficult to access real EVs datasets, limiting the scope and quality of data. This could result in insufficient model training or an overreliance on simulated data, impacting the model's generalizability and adaptability to emerging charging technologies and strategies.

Under the support of onboard chips, a deeper understanding of user charging habits can be integrated into the BMS model, utilizing the relatively stable data from electric vehicles during the charging process. Specifically, by leveraging fundamental data transmitted over the CAN bus, such as voltage, temperature, and time, it's possible to conduct detailed feature extraction of user charging behaviors, thereby optimizing BMS performance and enhancing user experience.

7. Conclusion

This paper focuses on the accurate estimation of SOH for LIBs. A novel SOH estimation method via feature optimization and CNN is proposed. Firstly, to describe LIBs' aging process more comprehensively, multiple features are extracted from Vt, DT, and IC curves considering electrical, thermodynamic, and electrochemical properties. Then the cross-validation's mean-variance is regarded as the objective function for searching the optimal feature dimension using SA. Feature optimization is realized based on PCA with the optimal feature dimension. The optimized features are used as CNN input to achieve accurate SOH estimation. Finally, we have conducted feature and model validation experiments to verify the proposed method's effectiveness. The results show that multiple features outperform single features in terms of the description of battery aging. Compared with the CNN and PCA-CNN models, the SA-PCA-CNN model has higher accuracy. Specifically, within the Oxford dataset, the SA-PCA-CNN model demonstrated an average MAE of 0.24%, which is significantly lower than that of the CNN model at 0.72% and the PCA-CNN model at 0.46%. Additionally, the average RMSE for the SA-PCA-CNN model stands at 0.31%, again surpassing the CNN model's 0.90% and the PCA-CNN model's 0.67%. This indicates that the SA-PCA-CNN model provides higher accuracy and lower error margins in estimating battery SOH. Furthermore, the estimation precision of SOH by the SA-PCA-CNN in our independent experimental dataset also exceeded that of the unoptimized CNN and PCA-CNN. Future work will focus on both datasets and algorithms. We aim to validate the performance of the proposed method on random charge-discharge datasets that align with real-world operating conditions. Additionally, incorporating transfer learning could further explore the effectiveness of the proposed method under different operating conditions.

CRediT authorship contribution statement

Mingqiang Lin: Writing – review & editing, Validation, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Leisi Ke:** Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Conceptualization. **Wei Wang:** Writing – review & editing, Validation, Software, Funding acquisition. **Jinhao Meng:** Writing – review & editing, Validation, Methodology, Funding acquisition, Conceptualization. **Yajuan Guan:** Writing – review

& editing, Validation, Methodology, Funding acquisition, Conceptualization. **Ji Wu:** Writing – review & editing, Validation, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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